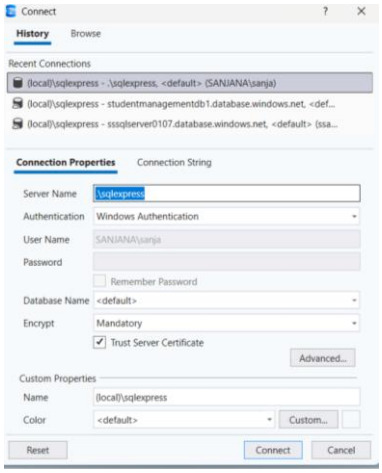
**DB First Approach**

1. Opened the SSMS, and connected it to the server .\sqlexpress




2. Search for NorthWind database that it is present in it having tables products, categories.
3. In visual studio, create a web api project called NorthWind.Services. In that install the packages in the developer powershell using commands
    - *Dotnet add package Microsoft.EntityFrameworkCore.SqlServer –version 8.0.10*
    - *Dotnet add package Microsoft.EntityFrameworkCore.Design –version 8.0.10*
    - *Dotnet add package Microsoft.EntityFrameworkCore.Tools –version 8.0.10*
4. In the project -> right click -> manage nuGet packages -> search for Automapper -> install.
5. Search for AutoMapper.Extensions.Microsoft.DependencyInjection -> install
6. Make a folder named Data. Then, In the package manager console, write this command for scaffolding dbcontext and models
   *dotnet ef dbcontext scaffold "Server=.\sqlexpress;Database=Northwind;Trusted_Connection=True;TrustServerCertificate=True;" Microsoft.EntityFrameworkCore.SqlServer -o Models -c NorthWindContext --context-dir Data -t Categories -t Products --use-data-annotations –force*
7. After running this command, we will find NorthwindContext.cs file in Data folder. And models folder will be automatically created.
8. In the NorthwindContext.cs file, override onModelCreating with this code

```
0 references
public NorthWindContext(DbContextOptions<NorthWindContext> options)
    : base(options)
{
}

0 references
public virtual DbSet<Category> Categories { get; set; }

0 references
public virtual DbSet<Product> Products { get; set; }

0 references
protected override void OnConfiguring(DbContextOptionsBuilder optionsBuilder)
#warning To protect potentially sensitive information in your connection string, you should move it out of source code. You can avoid sca
    => optionsBuilder.UseSqlServer("Server=.\\sqlexpress;Database=Northwind;Trusted_Connection=True;TrustServerCertificate=True;");
0 references
protected override void OnModelCreating(ModelBuilder modelBuilder)
{
    modelBuilder.Entity<Product>()
        .Property(p => p.UnitPrice)
        .HasColumnType("decimal(10,2)");

    modelBuilder.Entity<Product>()
        .HasOne(p => p.Category)
        .WithMany(c => c.Products)
        .HasForeignKey(p => p.CategoryId);
}
```

9. Create the DTOs folder, make three DTOs named CategoryDto, ProductDto, CategorySummaryDto.
10. Make a folder named Mappings and make a class called MappingProfile.

```
using AutoMapper;
using NorthWind.Services.Models;
using NorthWind.Services.DTOs;

namespace NorthWind.Services.Mappings
{
    1 reference
    public class MappingProfile : Profile
    {
        0 references
        public MappingProfile()
        {
            CreateMap<Category, CategoryDto>()
                .ForMember(dest => dest.ImageUrl,
                    opt => opt.MapFrom(src => "/images/" + src.CategoryName + ".jpg"));

            CreateMap<Product, ProductDto>();
        }

    }
}
```

11. Make all the controllers – CategoriesController

```
using AutoMapper;
using Microsoft.AspNetCore.Http;
using Microsoft.AspNetCore.Mvc;
using NorthWind.Services.DTOs;
using NorthWind.Services.Repository;

namespace NorthWind.Services.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    1 reference
    public class CategoriesController : ControllerBase
    {
        private readonly ICategoryRepository _repo;
        private readonly IMapper _mapper;

        0 references
        public CategoriesController(ICategoryRepository repo, IMapper mapper)
        {
            _repo = repo;
            _mapper = mapper;
        }
        [HttpGet]
        0 references
        public async Task<IActionResult> Get()
        {
            var categories = await _repo.GetAllAsync();
            return Ok(_mapper.Map<IEnumerable<CategoryDto>>(categories));
        }

    }
}
```

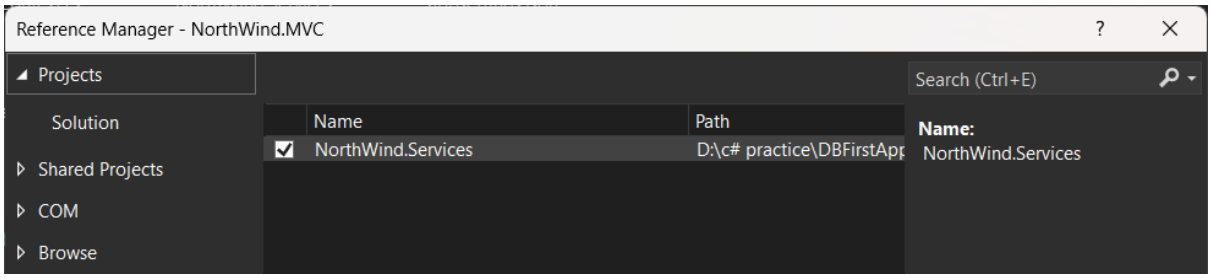
ProductsController

```
using AutoMapper;
using Microsoft.AspNetCore.Http;
using Microsoft.AspNetCore.Mvc;
using NorthWind.Services.DTOs;
using NorthWind.Services.Repository;

namespace NorthWind.Services.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    1 reference
    public class ProductsController : ControllerBase
    {
        private readonly IProductRepository _repo;
        private readonly IMapper _mapper;

        0 references
        public ProductsController(IProductRepository repo, IMapper mapper)
        {
            _repo = repo;
            _mapper = mapper;
        }

        [HttpGet("by-category/{categoryId}")]
        0 references
        public async Task<IActionResult> GetByCategory(int categoryId)
        {
            var products = await _repo.GetByCategoryIdAsync(categoryId);
            return Ok(_mapper.Map<IEnumerable<ProductDto>>(products));
        }

    }
}
```

SummaryController

```
using AutoMapper;
using Microsoft.AspNetCore.Http;
using Microsoft.AspNetCore.Mvc;
using NorthWind.Services.Repository;

namespace NorthWind.Services.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    1 reference
    public class SummaryController : ControllerBase
    {
        private readonly IProductRepository _repo;
        private readonly IMapper _mapper;

        0 references
        public SummaryController(IProductRepository repo, IMapper mapper)
        {
            _repo = repo;
            _mapper = mapper;
        }
        [HttpGet("summary")]
        0 references
        public async Task<IActionResult> GetSummary()
        {
            var data = await _repo.GetCategorySummariesAsync();
            return Ok(data);
        }

    }
}
```

12. Make a new Project **Northwind.MVC** and add a project reference of **Northwind.Services** to access the DTOs present in Web API.



13. Make the three controllers given for MVC i.e. ProductsController, SummaryController and CategoriesController.

-ProductsController

```
using Microsoft.AspNetCore.Mvc;
using NorthWind.Services.DTOs;

namespace NorthWind.MVC.Controllers
{
    1 reference
    public class ProductsController : Controller
    {
        private readonly HttpClient _client;

        0 references
        public ProductsController(IHttpClientFactory clientFactory)
        {
            _client = clientFactory.CreateClient("API");
        }
        0 references
        public IActionResult Index()
        {
            return View();
        }
        0 references
        public async Task<IActionResult> ByCategory(int id)
        {
            var products = await _client.GetFromJsonAsync<List<ProductDto>>
                ($"api/products/by-category/{id}");

            return View(products);
        }
```

-CategoriesController

```
using Microsoft.AspNetCore.Mvc;
using NorthWind.Services.DTOs;

namespace NorthWind.MVC.Controllers
{
    1 reference
    public class CategoriesController : Controller
    {
        private readonly HttpClient _client;

        0 references
        public CategoriesController(IHttpClientFactory clientFactory)
        {
            _client = clientFactory.CreateClient("API");
        }

        0 references
        public async Task<IActionResult> Index()
        {
            var data = await _client.GetFromJsonAsync<List<CategoryDto>>("api/categories");
            return View(data);
        }
    }
```

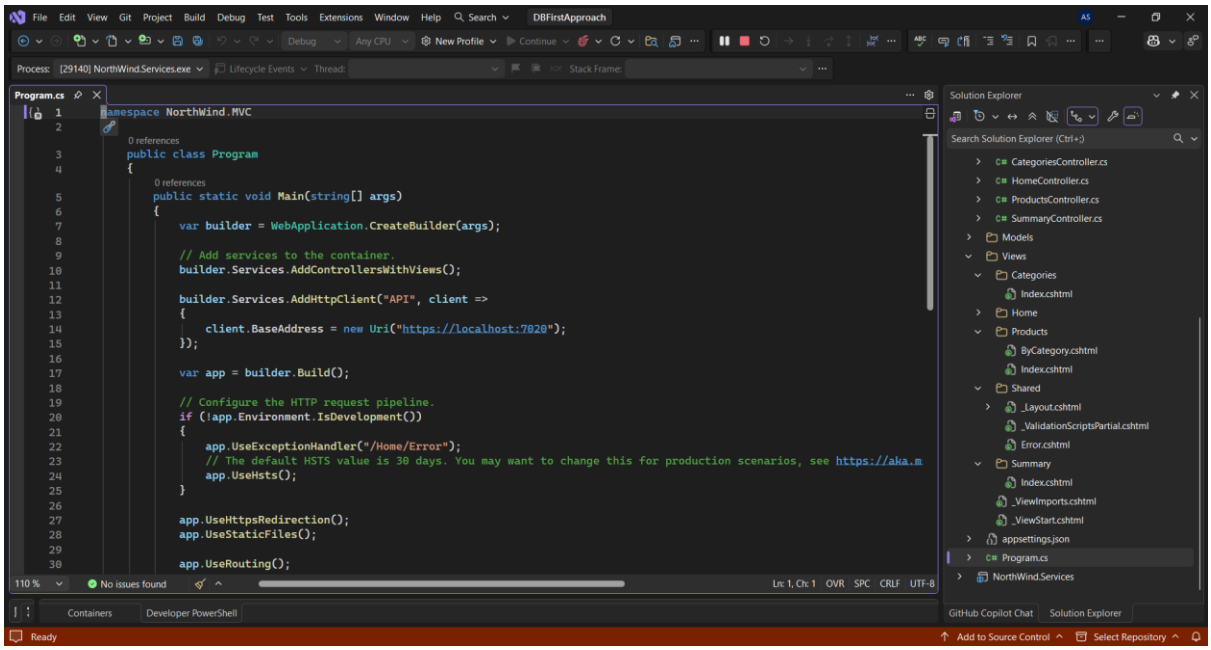
-SummaryController

```
using Microsoft.AspNetCore.Mvc;
using NorthWind.Services.DTOs;

namespace NorthWind.MVC.Controllers
{
    1 reference
    public class SummaryController : Controller
    {
        private readonly HttpClient _client;

        0 references
        public SummaryController(IHttpClientFactory clientFactory)
        {
            _client = clientFactory.CreateClient("API");
        }
        0 references
        public async Task<IActionResult> Index()
        {
            var summary = await _client.GetFromJsonAsync<List<CategorySummaryDto>>
                ("api/products/summary");

            return View(summary);
        }
```

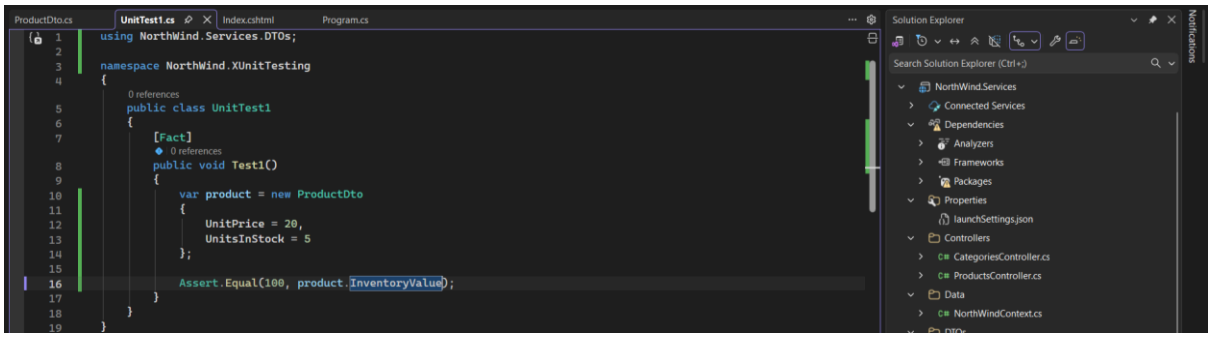
14. After making the controllers, make views for every action methods define in the controller.
15. In the appSettings.json file, add the webApi url.

```
Schema: https://www.schemastore.org/appsettings.json
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*",
  "baseUrl": "https://localhost:7020"
```

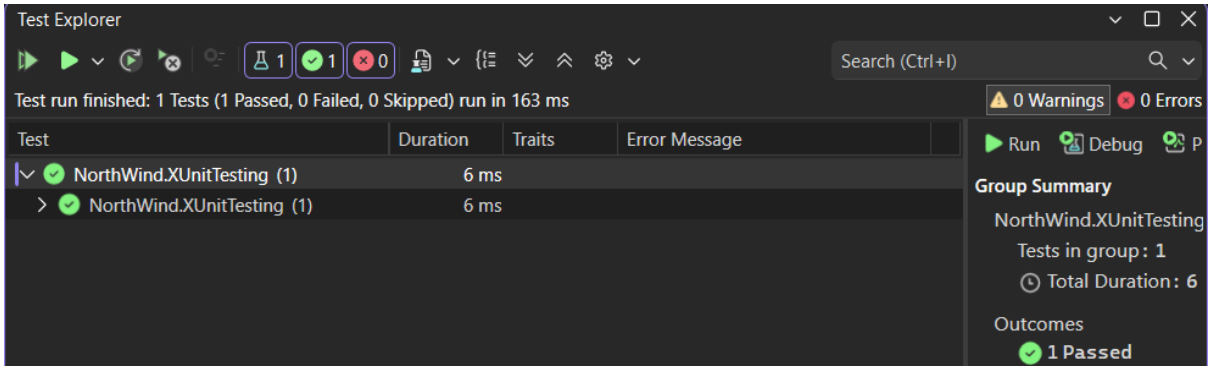
16. In the program.cs file, Add the Services of httpclient



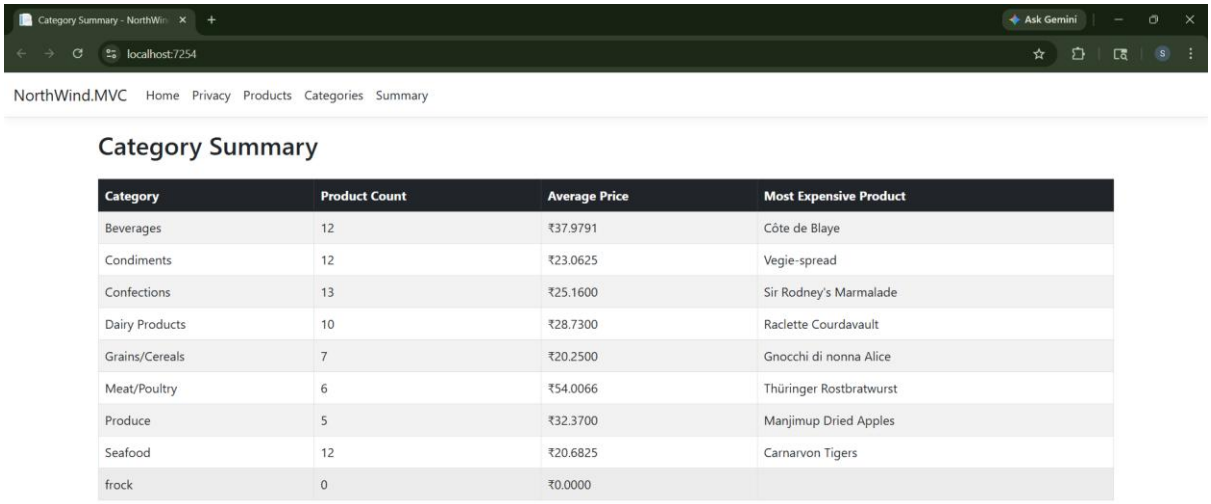

17. Add the buttons in the _layout.cshtml file for easy navigation.
18. Make another project of **XUnitTesting** in the Solution.
19. Again adding the project reference of NorthWind.Services to access data from it.
20. Adding one TestCase in it.




21. Checking in the test explorer that test cases are passed or not



22. In the mvc program.cs changing the default to Summary/Index. Then run it. It will show like shown below:




| Category | Product Count | Average Price | Most Expensive Product |
|---|---|---|---|
| Beverages | 12 | ₹37.9791 | Côte de Blaye |
| Condiments | 12 | ₹23.0625 | Vegie-spread |
| Confections | 13 | ₹25.1600 | Sir Rodney's Marmalade |
| Dairy Products | 10 | ₹28.7300 | Raclette Courdavault |
| Grains/Cereals | 7 | ₹20.2500 | Gnocchi di nonna Alice |
| Meat/Poultry | 6 | ₹54.0066 | Thüringer Rostbratwurst |
| Produce | 5 | ₹32.3700 | Manjimup Dried Apples |
| Seafood | 12 | ₹20.6825 | Carnarvon Tigers |
| frock | 0 | ₹0.0000 | |